

Dual-Domain MMD Adaptation: Large-Source, Small-Target Runs vs. a No-MMD Fine-Tune Control

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1 Setup

Both directions of adaptation were run from a large (full-size) pretrained checkpoint, fine-tuned against the small opposite-domain target set:

- **real_large_source_syn_small_target:** source = real_large (pretrained, detached, forward-only reference), target = syn_small (fine-tuned via detection loss, pulled toward source via MMD). 50 epochs.
- **syn_large_source_real_small_target:** mirror direction, source = syn_large, target = real_small. 50 epochs.

For each direction, a no-MMD control was run as a plain continued fine-tune from the same starting checkpoint, on the same small target set, with no MMD term and no source batches at all:

- **syn_small_from_real_large:** plain fine-tune of real_large’s checkpoint on syn_small only. 100 epochs.
- **real_small_from_syn_large:** plain fine-tune of syn_large’s checkpoint on real_small only. 100 epochs.

Exposure-count note. Epoch length in the MMD runs is defined by the *source* domain (the large set), with the small target resampled randomly each step; at batch size 8 this works out to roughly $200\times$ oversampling of the 1600-image target set over 50 epochs. The no-MMD controls, training directly on the small target set, get exactly $100\times$ exposure over their 100 epochs. The MMD runs therefore received *more*, not less, total exposure to the target set than their no-MMD counterparts – so the comparison below is not biased in the no-MMD control’s favor.

2 Results

3 Discussion

No visible MMD benefit on the target-domain class that matters most. On the metric of primary interest – vehicle mAP on whichever domain is actually being fine-tuned for – the MMD and no-MMD runs are essentially indistinguishable in both directions: 0.877/0.625 vs. 0.877/0.626 (real, direction 1) and 0.697/0.483 vs. 0.693/0.484 (syn, direction 2). The tiny edge that exists

Model	p mAP50	p mAP50-95	v mAP50	v mAP50-95
real_large_source_syn_small_target (MMD)	0.693	0.357	0.697	0.483
syn_small_from_real_large (no-MMD)	0.686	0.354	0.693	0.484
syn_large_source_real_small_target (MMD)	0.111	0.0266	0.214	0.138
real_small_from_syn_large (no-MMD)	0.129	0.0349	0.222	0.146

Table 1: Validated on the synthetic (syn) val set. Top pair: syn_small is the target domain (the direct MMD-vs-no-MMD comparison of interest). Bottom pair: syn is the source (forgotten) domain for that direction.

Model	p mAP50	p mAP50-95	v mAP50	v mAP50-95
syn_large_source_real_small_target (MMD)	0.642	0.296	0.877	0.625
real_small_from_syn_large (no-MMD)	0.644	0.303	0.877	0.626
real_large_source_syn_small_target (MMD)	0.028	0.00748	0.340	0.205
syn_small_from_real_large (no-MMD)	0.0146	0.00418	0.361	0.219

Table 2: Validated on the real val set. Top pair: real_small is the target domain (the direct comparison of interest). Bottom pair: real is the source (forgotten) domain for that direction.

flips sign between the two directions (no-MMD marginally ahead in direction 1, MMD marginally ahead in direction 2), consistent with run-to-run noise rather than a systematic effect.

This contrasts with the small-source/large-target regime. In that earlier setup (see `experiment_summary_v3.tex`), MMD showed a small but consistent vehicle-mAP improvement over the target domain’s own in-domain baseline, in both directions. Here, with the roles reversed (large source, small target), that benefit does not appear.

A plausible explanation. MMD’s contribution may be tied to how much direct, diverse target-domain data the model actually trains on. When the target set is large and gets extensive direct exposure (small-source/large-target regime), MMD’s feature-space regularization has room to meaningfully shape the shared backbone. When the target set is small and relies on heavy resampling with the same ~ 1600 images (large-source/small-target regime), plain fine-tuning on that same scarce set already extracts what there is to extract, and the added MMD machinery does not provide a further edge – despite the MMD runs here having received *more* total exposure to the target set (via oversampling) than the no-MMD controls, not less.

Caveats. These are single runs on val sets of modest size (160–640 images); the MMD runs used 50 epochs against the no-MMD controls’ 100, and while the oversampling arithmetic means MMD still saw more total target-image draws, epoch count and optimizer-schedule details (learning-rate decay shape in particular) were not otherwise matched. A repeat run, or matching total step count exactly rather than just approximately, would be needed before treating “no MMD benefit in this regime” as a settled result.